ISA AND CORE2350B (RP2350B)

SAMPLE RAM: on the module

The Core2350B2 has 8 MB of PSRAM on the RP2350B's QMI (CS1 = GPIO47, memory mapped). It replaces v2.0's U8 (APS6404L on GPIO0–3, PIO SPI); GPIO0–3 are now free.

I2S DAC

POWER

IRQ/DMA SETTING

MISC LEVEL SHIFT/BUFFER

AEN IO MASKING

Disables IOR and IOW signals when other DMA in system happens

ADDR/DATA MUX/DEMUX

RESET CIRCUIT

Latches BUSOE high after ADS is toggled for the first time, sets low on RESET

I/OCHRDY OPEN DRAIN

MOUNTING

74AHCxx LOGIC PWR

DACK INVERT

Saves PIO instructions due to JMP limitations

DECOUPLING CAPS

MIDI/DEBUG OUT

U6 74AHCT126 on +5V: OE high pulls the output low, OE low leaves it open (replaces the 74LVC2G06). Spare gates: A and OE tied low.

BOOT ROM (experimental)

Boot ROM

File: bootrom.kicad_sch

SST39SF010A in a PLCC-32 socket + 74HCT688: uses the global nets A0–A19, D0–D7, ~SMEMR and ROM_EN (GPIO2).

RADIO MODULE 2: WiFi / Bluetooth (CYW43439)

gSPI as on the Pico W: data in, data out (470R) and the IRQ (10k) share one GPIO. GPIO40–45 are the RP2350B pins that are not 5 V tolerant: fine here, all 3.3 V. Antenna end on the bracket edge; RF keep-out on the board (see README).

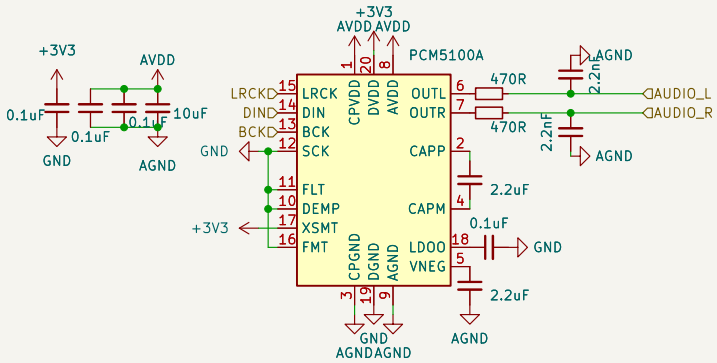
USB-C PROGRAM PORT

U12 module power: VSYS feeds the module's VBUS pin and its own LDO (ME6217, 3V3_EN pulled up on the module) powers the module. The module 3V3 pin is not connected: U9 powers the card logic. DAC (AVDD through FB4) and U13. GPIO30–47 are not used (GPIO39: module LED, GPIO47: module PSRAM CS on Core2350B1/B2).

5.1k Rd on CC1/CC2: device only, never sources VBUS. D+/D− shared with J5 (USB host) as on v2.0.

v2.0 ISA logic and GPIO map; TSSOP/SOIC-only chips as SMD
Through-hole PicoGUS 2.0 on a Waveshare Core2350B (RP2350B) module

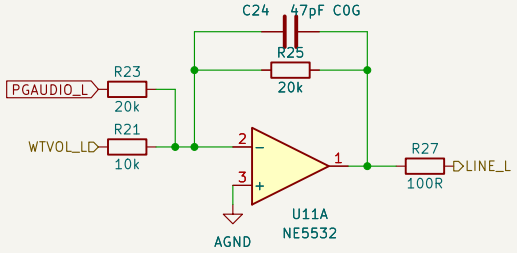
Sheet: /		
File: PicoGUS-THT.kicad_sch		
Title: PicoGUS v2.1 THT		
Size: A3	Date: 2026-10-29	**Rev: v2.1**
KiCad E.D.A. 10.0.7		Id: 1/4

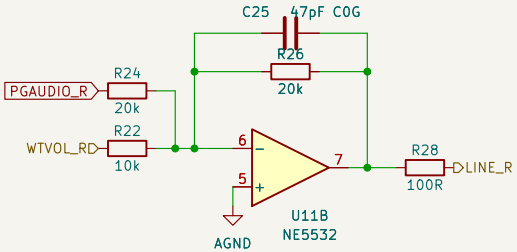


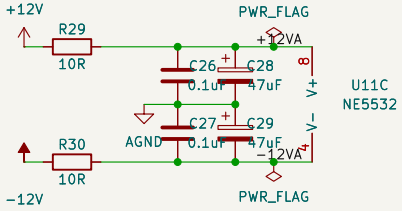
Sheet: /Audio/		
File: audio.kicad_sch		
Title: PicoGUS v2.1 THT – PCM5100A DAC		
Size: A4	Date: 2026-10-29	**Rev: v2.1**
KiCad E.D.A. 10.0.7		Id: 2/4

LINE-OUT MIXER

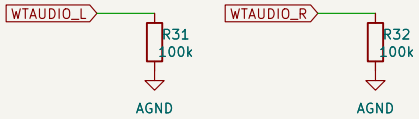
Inverting summing amplifier from THT v1.3: Waveblaster (after U13 M62429 volume) x2 = +6 dB through R21/R22 10k.
PCM5100A x1 through R23/R24 20k, feedback 20k || 47 pF (pole ~170 kHz). U11 runs from ISA +/-12V through 10R / 47uF.
The DAC output is ground-centred, so the PCM path is DC coupled (no electrolytic); the M62429 output sits at about
VCC/2, so C41/C42 (film) couple it. Both channels invert, so left and right keep their relative phase.







WT input pull-downs (THT v1.3 R14/R15)

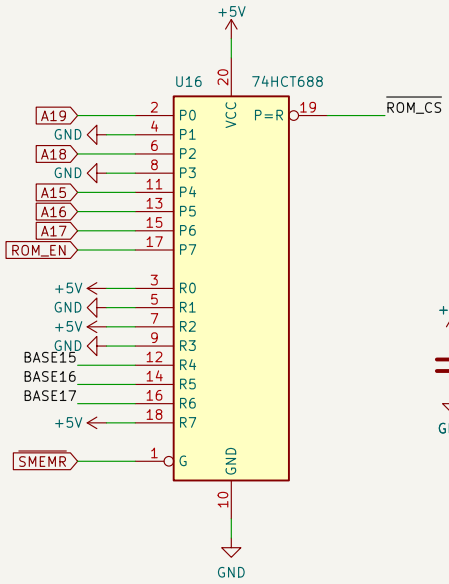


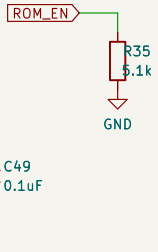
Sheet: /Mixer/		
File: mixer.kicad_sch		
Title: PicoGUS v2.1 THT – line-out mixer		
Size: A4	Date: 2026-10-29	**Rev: v2.1**
KiCad E.D.A. 10.0.7		Id: 3/4

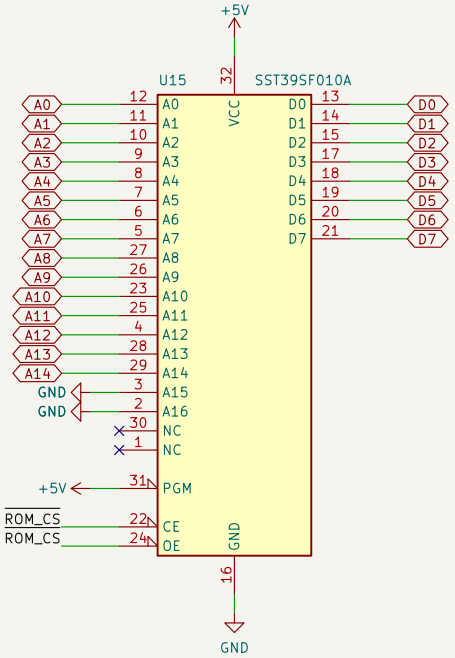
BOOT ROM: 32 KB option ROM on the ISA memory bus (experimental)

U16 compares SA19..SA15 with 1 1 JP3 JP2 JP1 and ROM_EN with 1 (GPIO2; R35 keeps the ROM off until the firmware enables it).

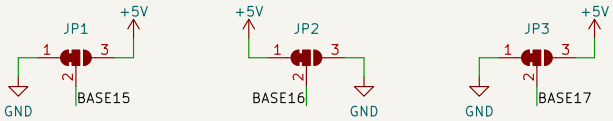
Its enable is ~SMEMR, so ~ROM_CS is low only during a memory read in the window: the flash (OE = CE) then drives the ISA data lines directly.











WE is tied high: program the flash in a programmer (it is in a socket).

Base address (JP3 JP2 JP1): 000 = C0000h, 001 = C8000h, 010 = D0000h (default), 011 = D8000h, 100 = E0000h, 101 = E8000h.

Each jumper: pads 1–2 are bridged by a track (cut it and bridge 2–3 to change the bit).

Sheet: /Boot ROM/		
File: bootrom.kicad_sch		
Title: PicoGUS v2.1 THT – boot ROM (experimental)		
Size: A4	Date: 2026-11-11	**Rev: v2.1**
KiCad E.D.A. 10.0.7		Id: 4/4